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Quiz 2

1 DP:

Advantages

- Guaranteed Convergence in the number of states
- Solves MDPs efficiently when the environment model is known

Weaknesses

- Needs complete knowledge of MDP transitions and rewards
- Curse of Dimensionality

2 MC:

Advantages

- Model-Free and Conceptual Simplicity
- Zero Bias

Weaknesses

- Can only be applied to episodic MDPs where all episodes must terminate
- High Variance

3 TD:

Advantages

- Model-Free like MC
- Handles Continuing Environments and Online Learning where MC can not
- Low Variance than MC

Weaknesses

- Introduces bias due to bootstrapping
- Sensitivity to Initial Values